

Finding brain oscillations with power dependencies in neuroimaging data

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ABSTRACT

Phase synchronization among neuronal oscillations within the same frequency band has been hypothesized to be a major mechanism for communication between different brain areas. On the other hand, cross-frequency communications are more flexible allowing interactions between oscillations with different frequencies. Among such cross-frequency interactions amplitude-to-amplitude interactions are of a special interest as they show how the strength of spatial synchronization in different neuronal populations relates to each other during a given task. While, previously, amplitude-to-amplitude correlations were studied primarily on the sensor level, we present a source separation approach using spatial filters which maximize the correlation between the envelopes of brain oscillations recorded with electro-/magnetoencephalography (EEG/MEG) or intracranial multichannel recordings. Our approach, which is called canonical source power correlation analysis (cSPoC), is thereby capable of extracting genuine brain oscillations solely based on their assumed coupling behavior even when the signal-to-noise ratio of the signals is low. In addition to using cSPoC for the analysis of cross-frequency interactions in the same subject, we show that it can also be utilized for studying amplitude dynamics of neuronal oscillations across subjects. We assess the performance of cSPoC in simulations as well as in three distinctively different analysis scenarios of real EEG data, each involving several subjects. In the simulations, cSPoC outperforms unsupervised state-of-the-art approaches. In the analysis of real EEG recordings, we demonstrate excellent unsupervised discovery of meaningful power-to-power couplings, within as well as across subjects and frequency bands.

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Introduction

The investigation of neural rhythms is a major aspect of the ongoing endeavor to understand the principles of brain functions (Buzsáki and Draguhn, 2004; Colgin et al., 2009; Womelsdorf and Fries, 2007). In the past decades, research on brain oscillations measured by electro-physiological (e.g., electro-/magnetoencephalography, EEG/MEG, and electrocorticography, ECoG) recordings has provided a large body of evidence suggesting a key role of neural rhythms in sensory processing

(Makeig and Jung, 1996; Thut et al., 2006), memory encoding (Jensen et al., 2007; Klimesch, 1999), and other cognitive functions (see reviews by Rieder et al. (2011) and Başar (2012), for example).

Interactions between oscillatory processes have been of particular interest in the recent past (Canolty and Knight, 2010; Engel et al., 2013; Hipp et al., 2012; Jensen and Colgin, 2007). Different types of cross-frequency coupling between neural systems have been identified and associated with specific brain functions. Such couplings can be divided into phase-to-frequency coupling, phase-to-phase coupling, phase-to-power coupling, and finally power-to-power coupling (e.g. Siegel et al., 2012). Here we focus on power-to-power (or amplitude-to-amplitude) coupling, which describes the interaction between the spectral power (or the envelope/amplitude) of distinct oscillations at specific frequencies or narrow frequency bands.

The power (or amplitude) of macroscopic EEG/MEG/ECoG or local field potential (LFP) oscillations is a function of the spatial

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synchronization strength in a given neuronal population (Denker et al., 2011; Pfurtscheller and Lopes da Silva, 1999), and thus a direct marker of neuronal activation in the brain regions generating these rhythms. Studying power-to-power coupling of distinct rhythms thus amounts to directly assessing the functional interactions between the associated brain sites.

Intra-subject power-to-power coupling may be observed as a (positive or negative) correlation of the power dynamics of two neural processes within the same frequency band or across frequency bands, and is hypothesized to mediate long as well as short range information transfer between brain areas (Fitzgerald et al., 2013; Furl et al., 2013). In social neuroscience studies investigating *inter*-subject interaction, power-to-power coupling between subjects can reveal leader/follower relations (Sänger et al., 2013). Additionally, the analysis of inter-subject power-to-power coupling could identify brain responses to potentially complex external stimuli purely based on the assumption that the stimulus-induced power dynamics are consistent across subjects. This is the idea of *hyperscanning* (Hasson et al., 2004; Montague et al., 2002).

Multivariate EEG/EMG scalp recordings are inherently ‘noisy’. Moreover, activity measured at a given sensor cannot be attributed only to the neuronal tissue directly beneath that sensor, but is generally a mixture of contributions from several neuronal sources, some of which might be far away from the sensor (Nunez, 2006, see Fig. 1 for an illustration). The problem of recovering the underlying neural signals from multivariate sensor readings is called (blind) source separation (BSS), and is inherently ill-posed, meaning that it relies on the presence of prior knowledge about these signals. The more precise the prior knowledge is, the better the true underlying brain processes can be recovered. For example, independent component analysis (ICA, see, e.g., Hyvärinen et al., 2001) assumes mutual statistical independence of the sources, and solves the BSS problem optimally if this assumption is met. If one is, however, interested in recovering sources exhibiting a specific type of dependency (among each other, or to an external variable) — such as power-to-power coupling of brain oscillations — it is reasonable to base the reconstruction of the source activity exactly on this assumed dependency rather than on mutual independence.

We have previously addressed the optimal extraction of neuronal oscillations the band-power of which is correlated with i) univariate target signals such as stimulus- or cognitive variables, and ii) multivariate target signals such as signals from a different measurement modality. Our work has led to the development of the source power correlation analysis (SPoC, Dähne et al., 2014) and the multimodal source power

correlation analysis (mSPoC, Dähne et al., 2013), which can also be applied to investigate phase-to-power coupling in electrophysiological recordings. The common philosophy behind SPoC and mSPoC is that both *directly optimize the quantity of interest* in the particular application they are designed for, namely the correlation between the band-power time course and the time course of a potentially related signal. By adhering to the linear generative model underlying electrophysiological recordings, both algorithms are able to reverse the mixing of local brain activity that is due to volume conduction in the head, and thereby to recover local brain oscillations with high signal-to-noise ratio (SNR).

In the present work, we continue this line of research, and present a novel unsupervised analysis method for the optimal extraction of pairs of neuronal sources the band-power dynamics of which are correlated with each other. We refer to this method as *canonical source power correlation analysis* (cSPoC). We demonstrate the utility of cSPoC in simulations as well as in three relevant application scenarios, each involving real EEG data of multiple subjects. These scenarios are:

- the isolation of oscillatory brain responses modulated by a complex auditory stimulus, based on the idea that the band-power dynamics of these responses are similar across multiple repetitions of the stimulus and for multiple subjects.
- the detection of task-related sensori-motor rhythmic (SMR) components in a brain–computer interface (BCI) setting, solely based on the consideration that the BCI paradigm induces anti-correlations of power dynamics.
- the blind recovery of pairs of alpha-band oscillations and their corresponding first higher harmonics in the beta-band, based on the idea that the various harmonics of the alpha rhythm are different aspects of the same neural process and should thus exhibit strong envelope correlations.

Note that for all three application scenarios considered the ground truth is either known or at least a strong hypothesis about the expected outcome exists. Therefore, all three analyses serve as real-world verifications for the effectiveness of cSPoC.

The remainder of this article is organized as follows. First, we recall the forward model that underlies electrophysiological recordings. Next, we propose cSPoC, a method designed to optimally extract sources exhibiting power-to-power correlations within this generative model. Thereafter, we present a simulation study, in which we compare cSPoC to relevant state-of-the-art approaches. Moreover, the experimental protocol and data analysis of three EEG studies is described.

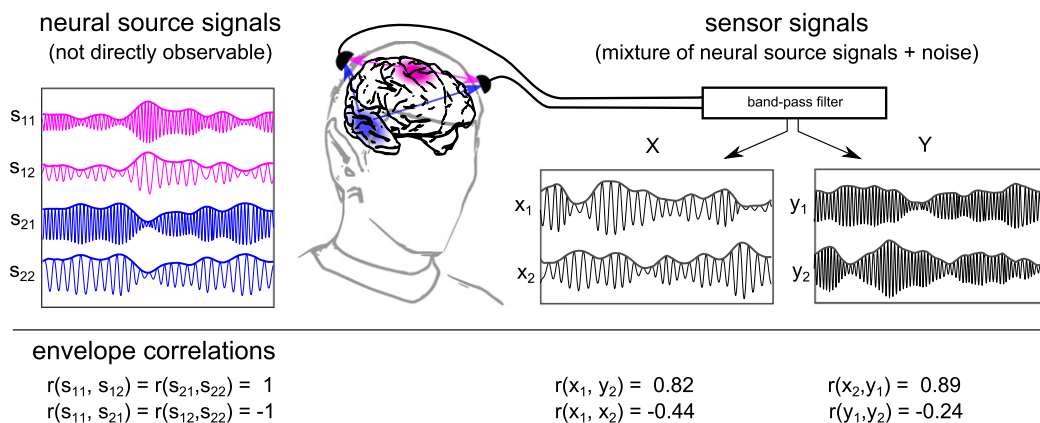


Fig. 1. Illustration of the problem setting. Left: Four hypothetical neural source signals that arise from two distinct brain regions (temporal lobe, shaded in blue, and frontal lobe, shaded in pink), where each brain region emits a fast and a slow oscillatory signal. The signals that correspond to the same brain region (e.g. s_{11} and s_{12}) share the same envelope, while the envelopes of the respective other neural source region have opposite envelope dynamics. Thus, in this illustration there is perfect envelope correlation across frequency bands (i.e. within brain regions) as well as perfect envelope anti-correlation within frequency bands (i.e. across brain regions). Right: Sensors on the scalp measure a linear superposition (weighted sum) of the neural source signals. In this example the fast and slow source signals were mixed separately with random mixing matrices and sensor noise. The envelope correlations of the sensor signals are strongly diminished compared to the envelope correlations of the source signals. Note that while the time courses of sensor signals are linear combinations of the source signal time courses, the envelopes of the sensor signals are *not* simply linear combinations of the source envelopes. Thus, linear projections of the envelopes of sensor signals cannot in general recover the source envelopes.

The results are presented and discussed before we conclude with some final remarks.

Methods

The generative model of electrophysiological recordings

Any analysis aiming at isolating the local electrical activity of single brain structures or networks from EEG, MEG, ECoG and LFP measurements must take into account that the generation of these measurements is governed by volume conduction in the various tissues in the head.

Let the measured data be represented by the multivariate variable $\mathbf{x}(t) \in \mathbb{R}^{N_x}$, where N_x denotes the number of sensors. Moreover, let $\mathbf{s}_x(t) \in \mathbb{R}^{K_x}$ denote the time courses of the electrical source activity of some underlying neural processes of interest. In electrophysiological recordings, the physics of volume conduction implies a linear mapping from these neural sources to the measurement (Baillet et al., 2001; Haufe et al., 2014; Parra et al., 2005). The generative (or forward) model thus reads

$$\mathbf{x}(t) = \mathbf{A}_x \mathbf{s}_x(t) + \boldsymbol{\varepsilon}_x(t), \quad (1)$$

where $\mathbf{A}_x \in \mathbb{R}^{N_x \times K_x}$ contains mixing coefficients describing the influence of the source activity on each sensor and $\boldsymbol{\varepsilon}_x(t) \in \mathbb{R}^{K_x}$ models noise contributions from outside the brain. The columns of $\mathbf{A}_x = (\mathbf{a}_x^1, \dots, \mathbf{a}_x^{K_x})$ are referred to as spatial activation patterns of the respective neural sources. Thus, $\mathbf{a}_x^i$, i.e. the i th column of $\mathbf{A}_x$, is the electrical potential or magnetic field pattern of the i th neural source. The matrix $\mathbf{A}_x$ can be regarded as the product of the so-called $N_x \times 3K_\beta$ lead field matrix $\mathbf{L}$ describing the EEG field spread of unit-magnitude electrical dipoles placed in a discrete grid or cortical mesh covering all possible source locations, and a $3K_\beta \times K_x$ matrix $\mathbf{J}$ reflecting the spatial distribution of each of the K_x components in the brain (Haufe et al., 2014). Estimating $\mathbf{J}$ given activation patterns $\mathbf{A}_x$ and a precomputed lead field $\mathbf{L}$ is called inverse source reconstruction, and allows one to map each of the components to actual brain anatomy (e.g., Baillet et al., 2001; Gramfort et al., 2013; Gross et al., 2001; Haufe et al., 2008, 2009, 2011). Note that, in reality, the number of neural sources K_β is much larger than the number of sensors N_x and hence larger as the maximum number of components K_x that can be estimated from the data using blind source separation techniques such as cSPoC, as explained below.

Canonical source power correlation analysis (cSPoC)

Here we want to find pairs of oscillatory neural sources from electrophysiological measurements based on the assumption that the envelope dynamics of these sources are correlated. Thus, we require the presence of a second set of measured data represented by the multivariate variable $\mathbf{y}(t) \in \mathbb{R}^{N_y}$ with sources $\mathbf{s}_y(t) \in \mathbb{R}^{K_y}$, mixing coefficients $\mathbf{A}_y \in \mathbb{R}^{N_y \times K_y}$ and noise $\boldsymbol{\varepsilon}_y(t) \in \mathbb{R}^{K_y}$. The datasets $\mathbf{x}$ and $\mathbf{y}$ are assumed to be related by $K < \min(K_x, K_y)$ pairs of source processes among the rows of $\mathbf{s}_x$ and $\mathbf{s}_y$, whose envelopes are linearly correlated. As a prerequisite for this, both datasets must be synchronized; that is, share a common time or sample index $t \in \{1, \dots, T\}$. This synchronization could be achieved by performing concurrent (e.g., EEG) recordings of two different subjects, by performing concurrent recordings of a single subject in two different measurement modalities (e.g., EEG and ECoG), by performing two recordings of a single subject (or single recordings of two different subjects) under identical stimulation, by creating two independent spectral representations of a single measurement through spectral filtering in two different frequency bands, or simply by using the same dataset for both $\mathbf{x}$ and $\mathbf{y}$, and so on.

A discriminative modeling approach

We are ultimately interested in the time-courses ($\mathbf{s}_x, \mathbf{s}_y$) and corresponding activation patterns ($\mathbf{a}_x, \mathbf{a}_y$) of power-coupled neural sources in the two datasets $\mathbf{x}$ and $\mathbf{y}$. However, estimating both the sources and their activation patterns jointly leads to difficult optimization problems. We can reduce the computational complexity by resorting to a so-called discriminative (or backward) modeling approach. Here, the time-courses of K neural sources are estimated by projecting the data linearly onto a set of spatial extraction filters $\mathbf{W}_x \in \mathbb{R}^{N_x \times K}$:

$$\hat{\mathbf{s}}_x = \mathbf{W}_x^\top \mathbf{x}(t). \quad (2)$$

In order to keep the notation simple, we will from here on omit the “^” from $\hat{\mathbf{s}}_x$. Thus, in the following we will use $\mathbf{s}_x$ (and $\mathbf{s}_y$) to refer to the result of Eq. (2), i.e. the estimated sources.

It can be shown that for each backward model of the form Eq. (2), a corresponding forward model of the form Eq. (1) exists, for which the noise $\boldsymbol{\varepsilon}_x(t)$ is uncorrelated to the sources $\mathbf{s}_x(t)$ (Haufe et al., 2014). The activation patterns of this forward model can be obtained from the spatial filters via the transformation

$$\mathbf{A}_x = \Sigma_x \mathbf{W}_x \Sigma_{s_x}^{-1}, \quad (3)$$

where Σ_x denotes the covariance matrix of the data $\mathbf{x}(t)$, and Σ_{s_x} is the covariance matrix of the extracted sources $\mathbf{s}_x(t)$ (Haufe et al., 2014).

Importantly, the extraction filters of backward models do not allow one to directly identify the strengths with which the power-coupled sources are expressed in each sensor, which would be a prerequisite for determining their anatomical origin and neurophysiological relevance (that is, for enabling “neurophysiological interpretation”). This is only possible for the activation patterns of forward models (Blankertz et al., 2011; Haufe et al., 2014). Moreover, it is only the activation patterns that can be subjected to source localization techniques (see above) in order to link cognitive functions to specific brain areas. However, by virtue of transformation (3), we can pursue a backward modeling approach, allowing us to conveniently parameterize the cost function solely in terms of the extraction filters (see next section), while being able to achieve neurophysiological interpretability and source localization through analysis of the activation patterns of the corresponding forward model. In fact, all topographical maps shown in this article are activation patterns obtained through Eq. (3). Note, however, that the time courses of the components correspond exactly to the time courses of the sources, thus even without performing a source modeling, we can directly extract and analyze corresponding time courses with cSPoC.

Extracting a single source pair

Let $\mathbf{w}_x \in \mathbb{R}^{N_x}$ and $\mathbf{w}_y \in \mathbb{R}^{N_y}$ be extraction filters for a single source pair $\mathbf{s}_x = \mathbf{w}_x^\top \mathbf{x}$ and $\mathbf{s}_y = \mathbf{w}_y^\top \mathbf{y}$. The instantaneous amplitudes (envelopes) of these sources are given by

$$\Phi[\mathbf{s}_x] = \sqrt{(\mathbf{w}_x^\top \mathbf{x})^2 + (\mathbf{w}_x^\top \mathbf{H}[\mathbf{x}])^2} \quad (4)$$

$$\Phi[\mathbf{s}_y] = \sqrt{(\mathbf{w}_y^\top \mathbf{y})^2 + (\mathbf{w}_y^\top \mathbf{H}[\mathbf{y}])^2}, \quad (5)$$

where $\mathbf{H}[\cdot]$ denotes the Hilbert transform (Barlow, 1993). We use Φ_x and Φ_y as shorthand notations for $\Phi[\mathbf{w}_x^\top \mathbf{x}]$ and $\Phi[\mathbf{w}_y^\top \mathbf{y}]$ from here on.

With these definitions, we can express the correlation between the source envelopes as a function of the spatial filters $\mathbf{w}_x$ and $\mathbf{w}_y$. The cSPoC objective function thus reads

$$\max_{\mathbf{w}_x, \mathbf{w}_y} c_{sg} \cdot \text{Corr}[\Phi_x, \Phi_y], \quad (6)$$

where the constant $c_{sg} \in \{+1, -1\}$ decodes whether positive or negative correlations are desired, and where

$$\begin{aligned} \text{Corr}[\Phi_x, \Phi_y] &= \frac{\text{Cov}[\Phi_x, \Phi_y]}{\sqrt{\text{Var}[\Phi_x] \text{Var}[\Phi_y]}} \\ &= \frac{\langle \bar{\Phi}_x \bar{\Phi}_y \rangle}{\sqrt{\langle (\bar{\Phi}_x)^2 \rangle \langle (\bar{\Phi}_y)^2 \rangle}} \end{aligned} \quad (7)$$

is Pearson's product-moment correlation coefficient with $\langle \cdot \rangle$ denoting average over time and with $\bar{\Phi}_x = \Phi_x - \langle \Phi_x \rangle$ and $\bar{\Phi}_y = \Phi_y - \langle \Phi_y \rangle$.

The objective stated in Eq. (6) is a non-convex higher-order nonlinear function of the coefficients $\mathbf{w}_x$ and $\mathbf{w}_y$. We propose to optimize it by means of standard nonlinear optimization techniques such as the limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) algorithm (Nocedal, 1980) implemented in MATLAB's (The Mathworks) `fminunc` routine. The gradient of Eq. (6) is given in Appendix A.

Extracting further source pairs

Once the first pair of spatial filters has been found, subsequent pairs can be obtained from the data using a deflation scheme. Firstly, we assume, without loss of generality, that the data have been “whitened”, i.e., transformed to be decorrelated and have unit variance. The whitening transformation is achieved by multiplying the data with the inverse of the matrix square root of their respective covariance matrices, $\Sigma_x^{-1/2}$ and $\Sigma_y^{-1/2}$. If $p - 1$ pairs of spatial filters have already been obtained from the whitened data and are collected in the sets $\mathbf{W}_x = \{\mathbf{w}_x^1, \dots, \mathbf{w}_x^{p-1}\}$ and $\mathbf{W}_y = \{\mathbf{w}_y^1, \dots, \mathbf{w}_y^{p-1}\}$, the p^{th} filter pair can be obtained by the following steps:

1. Project $\mathbf{x}$ and $\mathbf{y}$ into the spaces that are orthogonal to the spans of $\mathbf{W}_x$ and $\mathbf{W}_y$ (their *null spaces*), respectively. The required projection matrices $\mathbf{B}_x \in \mathbb{R}^{N_x \times (N_x - p + 1)}$ and $\mathbf{B}_y \in \mathbb{R}^{N_y \times (N_y - p + 1)}$ can be obtained from singular value decompositions of $\mathbf{W}_x$ and $\mathbf{W}_y$. This yields new data sets $\tilde{\mathbf{x}} = \mathbf{B}_x^T \mathbf{x}$ and $\tilde{\mathbf{y}} = \mathbf{B}_y^T \mathbf{y}$.
2. Optimize Eq. (6) with $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{y}}$ to obtain the spatial filter pair $\tilde{\mathbf{w}}_x^p \in \mathbb{R}^{N_x - p + 1}$ and $\tilde{\mathbf{w}}_y^p \in \mathbb{R}^{N_y - p + 1}$.
3. Project $\tilde{\mathbf{w}}_x^p$ and $\tilde{\mathbf{w}}_y^p$ back into the data spaces of $\mathbf{x}$ and $\mathbf{y}$ to obtain the p^{th} filter pair $\mathbf{w}_x^p = \mathbf{B}_x \tilde{\mathbf{w}}_x^p$ and $\mathbf{w}_y^p = \mathbf{B}_y \tilde{\mathbf{w}}_y^p$.

Once the desired number of spatial filters has been obtained, the filters can be projected into the original (unwhitened) data spaces of $\mathbf{x}$ and $\mathbf{y}$ by multiplying them with $\Sigma_x^{1/2}$ and $\Sigma_y^{1/2}$, respectively. The nature of the outlined deflation scheme limits the number of extracted pairs to $P \leq \min(N_x, N_y)$, where N_x and N_y are the dimensionality of $\mathbf{x}$ and $\mathbf{y}$ respectively. Importantly, the number of meaningful pairs of components extracted with cSPoC (e.g. 5–10), is in general similar to the number of components obtained in other techniques applied to EEG/MEG such as different ICA decompositions.

Extensions

The cSPoC framework can be naturally extended in various directions, which we outline in the following paragraphs. The derivations needed for actually implementing these extensions are found in Appendix B.

Log-transformation. The co-modulation of Gaussian distributed variables is completely described by Pearson's correlation coefficient

(Lee Rodgers and Nicewander, 1988). However, envelope and band-power signals are not Gaussian distributed, as noted for example in Freyer et al. (2009). Thus, in practice, a logarithmic transformation is often applied to make band-power and envelope signals more Gaussian (e.g., Hipp et al., 2012) and thereby make obtained correlation values more meaningful. In order to explicitly optimize the correlation between log-envelopes, the cSPoC objective function can be modified as outlined in the [Logarithm](#) section in Appendix B.

Averaging within epochs. In some scenarios it might be beneficial to average the envelope of the sources across short time intervals. We refer to these intervals as epochs and denote them by the index e . Possible reasons for averaging envelopes within epochs may include (i) suppression of noise, (ii) reduction of the sampling rate of the envelope, or (iii) the data may have an epoch (or trial) structure already and the envelope dynamics within trials are not the subject of study but common dynamics across trials are. In this case, the cost function has to be modified, such that Φ_x and Φ_y are to be correlated across the epoch index e rather than across the original time index t . This modification is described in the [Average within epochs](#) section in Appendix B.

Time-delayed coupling. The assumed coupling between envelopes may not necessarily be instantaneous. Therefore, several authors have modeled non-instantaneous coupling dynamics explicitly using temporal delay operators (Bießmann et al., 2010; Campi et al., 2013; Dähne et al., 2013). We adopt this idea by applying a temporal filter $\mathbf{w}_\tau \in \mathbb{R}^{N_\tau}$ to one or both of the envelopes, where N_τ is the number of time-lags considered (see the [Time-lagged envelope coupling](#) section in Appendix B). The temporal filter $\mathbf{w}_\tau$ is estimated jointly with the filters $\mathbf{w}_x$ and $\mathbf{w}_y$, and models temporal delays as well as convolutive interactions between envelopes.

Multiple data sets. Finally, cSPoC is easily extended to settings in which envelope coupling between $N > 2$ data sets is suspected. Let $\mathbf{x}^n$ denote the n^{th} data set for $n \in \{1, \dots, N\}$. In this case we seek a set of N weight vectors $\mathbf{w}^n$. One straightforward way of generalizing correlation coefficient to $N > 2$ data sets is to optimize the sum of pair-wise correlations. This approach, which has been proposed by Kettenring (1971) in the context of canonical correlation analysis (CCA) can be easily adopted also in cSPoC.

Practical details

In order to ensure meaningful results, care has to be taken in cases where the two data sets $\mathbf{x}$ and $\mathbf{y}$ stem from same measurement and cSPoC is set to maximize envelope correlations between the projections $\mathbf{w}_x^T \mathbf{x}$ and $\mathbf{w}_y^T \mathbf{y}$. If $\mathbf{x}$ and $\mathbf{y}$ are identical, the trivial solution to maximizing the objective stated in Eq. (6) is given by $\mathbf{w}_x = \mathbf{w}_y = \mathbf{w}$, which means that any pair of spatial filters will be optimal as long as they are identical. The trivial solution can be avoided by imposing additional constraints on $\mathbf{w}_x$ and $\mathbf{w}_y$ or, alternatively, by ensuring that $\mathbf{x}$ and $\mathbf{y}$ are sufficiently different. The latter can be achieved by band-pass filtering in different frequency bands, for example. Note that the problem of trivial solutions does not arise if $\mathbf{x}$ and $\mathbf{y}$ are identical and the correlation is to be minimized, i.e. envelope anti-correlations are investigated.

The objective function optimized by cSPoC is non-convex, which means that there is the potential for the optimization algorithm to end up in local optima. In practice, we overcome this issue by restarting the optimization with different, randomly selected initiations. We then use the solution that yielded the highest envelope correlation on the part of the data that was used to fit the model and apply it to the validation data. For current computers (depending on the amount of data and its dimensionality) the runtime for a single optimization is in the order of seconds and thus allows for multiple restarts. We have found 10 to 15 restarts to be sufficient for the algorithm to arrive at the same optimum several times.

A MATLAB implementation of cSPoC including all extensions outlined above is available upon request.

Relation to canonical correlation analysis

Our algorithm is called *canonical source power correlation analysis*, because it bears some conceptual similarity with canonical correlation analysis (CCA, Hotelling, 1936). CCA and its variants (Bießmann et al., 2010, 2011; Dmochowski et al., 2012) allow for the extraction of correlated components from two multivariate data sets. Both CCA and cSPoC seek linear projections of two data sets in order to maximize a correlation coefficient defined on the projected data (sources). However, in case of CCA, the correlation is defined between the source time courses directly, whereas in cSPoC the correlation is defined between nonlinear features of the sources, namely their envelopes. Importantly, CCA cannot be used to extract power-coupled brain sources from electrophysiological recordings. While it is conceivable to use a two-step approach, i.e., first compute the envelopes of the raw channel data of the two data sets and then apply CCA, such an approach does not lead to an accurate inversion of the generative model Eq. (1) (see also Dähne et al., 2013). The reason for this is that the two operations (i) source extraction through linear projection and (ii) nonlinear processing (computation of the envelope) do not commute. By design, cSPoC computes envelopes on the source level *after* projecting the data onto the extraction filters, and thus performs both operation in the order implied by the generative model. Fig. 2 illustrates graphically the fundamental difference between CCA and cSPoC.

Experiments

In this section, we describe an exhaustive simulation study designed to assess the performance of cSPoC as well as conventional approaches in a broad range of realistic scenarios. Moreover, we present three real-data studies well-suited to demonstrate cSPoC's effectiveness in recovering brain oscillations in a variety of practical applications.

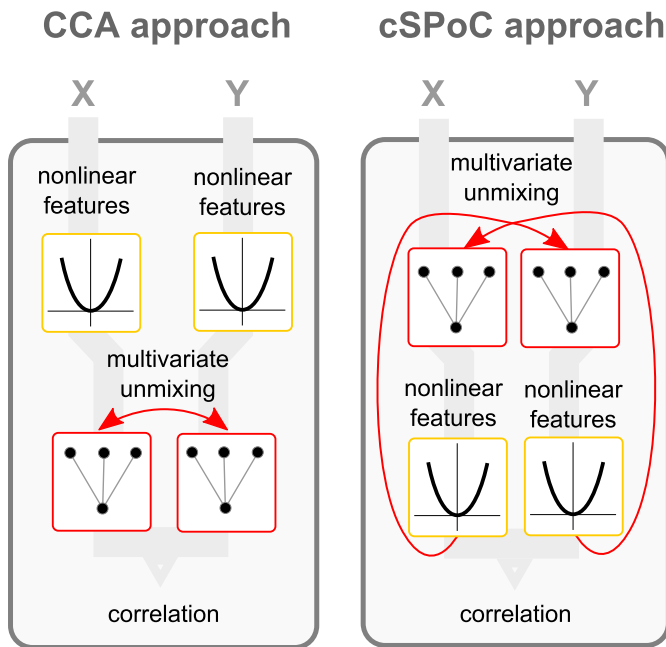


Fig. 2. Comparison of two multivariate approaches for power-to-power coupling analysis: canonical correlation analysis (CCA) and canonical source power correlation (cSPoC). Processing steps are to be read from top to bottom. Both methods receive the multivariate band-pass filtered data sets $\mathbf{x}$ and $\mathbf{y}$ as input. In case of CCA, there is first a channelwise computation of envelopes, which is a non-linear processing of the data. Thereafter, the multivariate envelope signals are projected to maximize correlations. cSPoC, on the other hand, searches for projections of the raw data which maximize the correlation of the envelopes of the projected signals. Since non-linear feature extraction (computation of envelope) and linear demixing do not commute, only cSPoC applies these operations in the order implied by the generative model of electrophysiological recordings and is therefore capable of recovering the actual power-coupled brain sources.

Simulations

Data generation

For each simulation, we created two artificial EEG datasets according to the generative model Eq. (1). The number of simulated channels was $N_x = N_y = 25$, and a total of $K_x = K_y = 31$ sources was simulated for each data set. The source time courses $\mathbf{s}_x$ and $\mathbf{s}_y$ were created by specifying an amplitude spectrum, in which the frequency bins of the alpha range (8 to 12 Hz) had non-zero values, while all other frequency bins were set to zero. The corresponding phase spectrum was drawn from a uniform distribution over the interval 0 to 2π . We applied the inverse Fourier transform to the combined amplitude and phase spectra in order to obtain band-limited oscillations with random envelope variations.

We simulated $K = 1$ pair of *target sources*, the envelopes of which were manipulated to exhibit a defined degree of correlation. The time courses (and envelopes) of all other sources were completely independent to each other and to the target sources. These independent sources are referred to as background sources. Target and background sources were separately projected to EEG electrodes using realistic activation patterns $\mathbf{A}_x$ and $\mathbf{A}_y$, which were obtained as the scalp potentials induced by a electrical dipoles at random brain locations and with random orientations in 3D voxel space. These potentials were calculated within a realistic head model (Fonov et al., 2011) using semi-analytic expansions of the electric lead fields (Nolte and Dassios, 2005). In a final step, target and background activity, as well as additional sensor noise were summed in sensor space to yield pseudo-EEG with a predefined signal-to-noise ratio (SNR). The equations describing the generation of the various signal components, as well as the exact definition of the SNR are given in Appendix C.

Tested methods

We benchmarked cSPoC against two other unsupervised methods, namely CCA and PowerCCA (Ramírez et al., 2013), as well as one supervised method, source power correlation analysis (SPoC, Dähne et al., 2014). CCA was applied as outlined in Fig. 2, i.e., envelopes were computed for each simulated channel, and the envelope time courses were used as input for CCA. All other methods act on the unprocessed time-domain data, and envelopes were computed after performing the backward projection, i.e., on the extracted sources. PowerCCA is a recent method that also optimizes spatial filters for two data sets such that the envelopes of the projected signals exhibit maximal correlation. We discuss the relation between CCA, PowerCCA and cSPoC in the Discussion section.

Since the “ground truth” (i.e., the time-courses and envelopes of the coupled sources) is known in this simulation setting, it is possible to apply supervised methods for extracting the sources-of-interest as baseline methods. Here, we applied SPoC, which seeks sources (one at a time), whose envelopes maximally correlate with a given target function. In order to recover pairs of coupled sources with SPoC, the algorithm was applied to data sets $\mathbf{x}$ and $\mathbf{y}$ separately, where the target function was always defined to be the known envelope of the matching simulated source. By using this knowledge, SPoC has a considerable advantage over cSPoC, CCA and PowerCCA, and therefore mainly serves as an empirical upper performance bound in the benchmarking.

Testing protocol

To compare the robustness of the tested methods, we systematically varied two parameters: (i) the strength of the correlation between the envelopes of the simulated target sources, and (ii) the signal-to-noise ratio (SNR). For each choice of target source correlation coefficient and SNR, 100 (pairs of) independent datasets were randomly drawn. Each dataset consisted of 15 min of data, of which the first 5 min¹ were

¹ Note that approximately 2 min of data are sufficient to optimize the parameters of all methods such that they reach their respective performance plateaus on the validation data.

used to fit the parameters of the algorithms (i.e. to optimize the spatial filters), while the remaining 10 min were used to validate them.

The correlation of the envelopes of the extracted sources, calculated on validation data, and averaged across the 100 repetitions of the experiments, served as the evaluation metric.

Identifying SSAEP generators based on inter-subject envelope correlations during identical stimulation

As an initial real-world example, we here consider the so-called *hyperscanning* setting, where envelope coupling of brain oscillations arises, because multiple subjects are stimulated in the same way. In the absence of any other information, this coupling can be utilized by cSPoC to extract the underlying oscillatory processes. Importantly, this can be done even if neither the stimulus sequence nor the actual relationship between stimulus and response is known.

Experimental paradigm

We consider EEG data of $N = 7$ healthy subjects, which had been recorded by Dähne et al. (2014). Subjects were exposed to 15 min (three blocks of 5 min) of ongoing stimulation with an auditory steady-state stimulus. This kind of stimulus is known to elicit a steady-state auditory evoked potential (SSAEP), which is visible as a spectral peak in the EEG at the steady-state stimulation frequency (Galambos et al., 1981; Hari et al., 1989; Picton et al., 2003). The loudness of the stimulus was slowly modulated. Since the magnitude of the spectral response is known to be positively linearly correlated with loudness (measured on a dB scale) (Plourde et al., 1991; Rodriguez et al., 1986), this induced corresponding envelope dynamics in the EEG at the steady-state frequency. By using the known stimulus intensity as the target function for SPoC, Dähne et al. could extract the SSAEP component from the ongoing EEG with high precision.

Data acquisition and preprocessing

EEG signals were recorded using a Fast'n Easy Cap (EasyCap GmbH) with 63 Ag/AgCl electrodes placed at symmetrical positions based on the international 10–20 system. Channels were referenced to the nose. Electrooculographic (EOG) signals were recorded in addition but not used in the present context. Signals were amplified using two 32-channel amplifiers (Brain Products) and sampled at 1 kHz.

We applied spatio-spectral decomposition (SSD, Nikulin et al., 2011) to perform dimensionality reduction by projecting the data onto the space spanned by the 10 components showing largest spectral peaks in the frequency band of interest, i.e., around 40 Hz. The SSD-projected data were then band-pass filtered using a pass-band ranging from 39 to 41 Hz.

The actual loudness modulations were not identical for all subjects in the original experiment. However, all modulation functions had the same uniform histogram over the presented loudness interval, which means that all subjects were exposed to the same intensity levels for the same number of times, only in a different order. Thus it is possible to create a *virtual* same-stimulus scenario from these data. In order to do so, the band-pass filtered data as well as the corresponding loudness modulation were segmented into consecutive non-overlapping epochs of 1 second length. For each pair of subjects S_i and S_j , the epochs of subject S_j were re-ordered such that the re-ordered loudness function of subject S_j became aligned with true loudness modulation function of subject S_i . Averaging both loudness functions within epochs and correlating the results yielded correlations between loudness functions that were larger than 0.99 for all subject pairs.

Application of cSPoC and SPoC

The SPoC method used in Dähne et al. (2014) is a supervised method relying (i) on knowledge of the stimulation intensity (in this case loudness) and (ii) on the assumption of a *linear* relationship between stimulus intensity and the envelope of the oscillatory brain response. While both prerequisites are fulfilled for the considered SSAEP data, this might not be the case in general. Here we show that competitive performance

can be obtained using cSPoC in a hyperscanning setting, i.e., when multiple datasets with identical stimulation are available. Notably, cSPoC is an unsupervised method neither requiring any information about the stimulation, nor assuming a specific (e.g., linear) relationship between stimulus and brain response.

Since we could not expect between-subject EEG envelope correlations *within* epochs but only *across* epochs, we applied the cSPoC version that optimizes correlations across epochs as introduced in the *Canonical source power correlation analysis (cSPoC)* section.

We applied cSPoC to all pairwise combinations of the data of the seven subjects. For each subject pair, only the first source pair (with highest envelope correlation) was extracted. SPoC, on the other hand, was applied to the data of each subject separately, using the corresponding loudness modulation as the target function, as done in Dähne et al. (2014). Due to this knowledge, SPoC has a crucial advantage over cSPoC.

The performance of both algorithms is measured in terms of the correlation of the envelopes of the extracted sources with the corresponding target variables, as well as by computing all possible pairwise correlations between the source envelopes obtained from different subjects. In order to exclude overfitting effects, we report average correlations obtained in a 10-fold cross-validation scheme.

Identifying generators of sensori-motor rhythms based on envelope anti-correlations in a BCI paradigm

In the second real-world example, we demonstrate how cSPoC can be used to find competing, or even mutually exclusive, cognitive components by looking for oscillatory EEG sources with anti-correlated envelopes within subjects. Since anti-correlations cannot emerge trivially from using identical projection filters, pairs of such sources can be found even within a single dataset.

Experimental paradigm, data acquisition and preprocessing

We consider data from a brain–computer interface (BCI) study with $N = 40$ subjects, in which the subjects had to alternately imagine movements of either their left hand, their right hand, or their feet. The data were originally collected for a large-scale screening study on BCI aptitude, described in Blankertz et al. (2010). Imagined movements of body parts leads to event-related desynchronization (ERD) of spatially distinct generators of sensori-motor rhythmic (SMR) activity, namely those linked to the imagined body parts. These localized ERD phenomena are the basis for BCI classification.

EEG signals were recorded using a Fast'n Easy Cap (EasyCap GmbH) with 118 Ag/AgCl electrodes placed at symmetrical positions based on the international 10–20 system. For each subject, a *calibration* and a *feedback* session was recorded. In the calibration session, subjects were instructed to perform motor imagery of their left hand (L), their right hand (R) or both of their feet (F) in randomized order after presentation of a respective cue. A total of 225 trials (75 per class, 3 classes) of 5 s length were recorded. Only the trials from the best discriminable pair of classes among the three possible class pairs (L vs. R, L vs. F, and R vs. F) were chosen for further analysis, yielding a total of 150 trials for each subject. For the selected pair of classes, a best-discriminating frequency band in the SMR range (5–30 Hz) and time interval relative to the cue was determined heuristically (Blankertz et al., 2008). SSD was used for dimensionality reduction and set to extract a 10 dimensional subspace that captures oscillatory activity in the subject-specific SMR frequency range.

Application of cSPoC and CSP

In BCI contexts, where class label information is known during the training phase of the system, common spatial patterns (CSP, e.g. Blankertz et al., 2008) analysis is typically applied to extract the sources exhibiting class-specific ERDs. CSP is a supervised decomposition technique seeking sources for which the ratio of the band-power in the

two classes (and thus, the ERD) is extremal. Here, we compare cSPoC to cSPoC, which does not use class label information at all, and thus has a disadvantage compared to CSP. However, considering that the BCI paradigm requires the various motor imagery tasks to be performed one at a time, anti-correlations of the envelopes of the sensori-motor rhythms associated with the imagined body parts should emerge. These anti-correlations enable cSPoC to identify the class-related ERD sources in a data-driven, purely unsupervised, manner.

Classification

As is common practice in SMR BCI, we used the logarithm of the band-power of the first six components extracted by the methods (i.e., the three best-correlating source pairs for cSPoC as obtained via the outlined deflation scheme, and the six components with the largest power ratios for CSP) as features for regularized linear discriminant analysis (RLDA) classifiers, which were trained using the true class labels.

Classification results were evaluated on data from the feedback session, in which the subjects were asked to perform 300 additional trials of motor imagery. Here, only the two classes providing best discrimination on the calibration data were used (150 trials each). The data were processed in the same way as the calibration data, using the same band-pass filters, time intervals, cSPoC/CSP spatial filters and classifier coefficients. For each of the two projection techniques (cSPoC and CSP), the classification accuracy (i.e., the percentage of correctly classified trials) was assessed.

Analyzing alpha and beta oscillations with coupled envelopes during rest

Cross- and within frequency interactions are hypothesized to be an important mechanism for integration of neuronal processes distributed across different brain areas and frequency bands. In this experiment, we focused on envelope interactions between and within alpha and beta oscillations during rest.

Previous studies, addressing a relationship between alpha and beta oscillations, primarily considered $n:m$ cross-frequency phase synchronizations (Carlqvist et al., 2005; Nikulin and Brismar, 2006; Palva et al., 2005). However, the presence of phase synchronization per se does not indicate that the envelopes of oscillations should correlate as well (Bayraktaroglu et al., 2013). Studies focusing on envelope dynamics within frequency bands reported positive correlations between envelopes of oscillations in the alpha and beta range, which were localized in sensory, as well as somatosensory networks (Engel et al., 2013; Hipp et al., 2012; Nikouline et al., 2001; Siegel et al., 2012).

Experimental paradigm, data acquisition and preprocessing

Seven subjects participated in the study (2 females). EEG recordings were performed at rest condition (15 min) with subjects seated comfortably with their eyes being open. EEG data were recorded with 96 Ag/AgCl electrodes, using BrainAmp amplifiers and BrainVision Recorder software (Brain Products GmbH, Munich, Germany). The signals were recorded in a 0.016–250 Hz frequency range and digitized at 1000 Hz. For the following offline analysis, the EEG data were decimated to 200 Hz. For dimensionality reduction, the data were projected to two separate 10 dimensional subspaces: the first SSD subspace captured oscillatory activity in the alpha range (9 to 12 Hz) and the second SSD subspace captured oscillatory activity in the beta range (19 to 23 Hz).

Application of cSPoC

The SSD-projected data were band-pass filtered separately in the alpha range (9 to 12 Hz) and in the beta range (19 to 23 Hz), yielding two 15 min long narrow-band multi-variate time-series per subject. For each subject, cSPoC received the alpha and beta band-passed data as input and five spatial filter sets were optimized using the deflation scheme outlined in the Methods section. Envelopes of the spatially filtered signals were averaged within consecutive, non-overlapping

windows of one second length and the resulting envelope time-courses were correlated. The entire analysis was carried out in a cross-validation loop (10 folds) to avoid overfitting.

In addition to investigating these cross-frequency envelope dynamics, we also assessed the envelope correlations *within* the alpha and beta bands. These were assessed using the first two cSPoC component pairs. For this test, we applied cSPoC to the entire data, i.e. without splitting in separate training- and validation data.² The envelope of the first alpha component (alpha 1) was correlated with the envelope of the second alpha component (alpha 2) similarly for the beta components). Note that envelope correlations between alpha (beta) components cannot be attributed to trivial interaction due to volume conduction. The deflation scheme of our optimization ensures orthogonality (i.e. decorrelation) between the time courses of alpha (beta) components.

Statistical significance of the envelope correlation was tested non-parametrically by a permutation approach: one of the envelope time-series was shuffled and correlations were computed for 1000 separate shuffles. The corresponding p-value was computed as the number of times in which the correlations obtained for the shuffled envelope exceeded the unshuffled correlation, divided by the total number of shuffles. The shuffling was implemented by first computing the Fourier spectrum (amplitude and phase) of one of the envelopes, then randomizing the phases and finally applying the inverse Fourier transform to the original amplitude spectrum. This approach ensures that the shuffled envelope has the same auto-correlation as the unshuffled envelope.

Results

Simulations

The results of the simulations are shown in Fig. 3. The figure depicts the correlations (obtained on validation data) between the envelopes of the extracted sources from simulated data sets $\mathbf{x}$ and $\mathbf{y}$. We show the recovered correlations as a function of the SNR for two different values (0.7 and 1) of the correlation of the envelopes of the simulated sources (the “true” correlation), as well as of the “true” correlation for two different SNRs (−5 dB and 0 dB).

All methods approach the true correlation between the target sources as the SNR increases above 5 dB. cSPoC attains the best performance among all unsupervised methods and is on par with the supervised SPoC over large SNR ranges. Note the large gap in performance between CCA and its competitors. CCA, by design, is able to invert linear mixtures of correlating time series. However, if the linear model (Eq. (1)) holds for the raw data (as it is the case for EEG data), this implies that the envelopes of the sources are *not* linearly mixed in the channel envelopes, making it difficult for CCA to extract the target envelopes. CCA is able to perform well only in high SNR regimes, when the target sources dominate the simulated EEG signal.

All methods eventually fail as the SNR becomes very low (i.e., below −5 dB) and/or the true correlation approaches zero. However, we observe that cSPoC outperforms its unsupervised peers across wide parameter regimes. cSPoC outperforms PowerCCA with statistical significance in the range of −7 dB and 10 dB and for all correlations larger or equal than 0.3 (paired t-test, $p < 0.01$).

Identifying SSAEP generators based on inter-subject envelope correlations during identical stimulation

Fig. 4 shows cross-validated correlation coefficients obtained by SPoC and cSPoC in the re-analysis of the auditory steady-state data. The subjects are ordered by the correlations between stimulus intensity and envelope as obtained with SPoC, from highest to lowest. Plot A in

² This does not represent a confound because cSPoC was used to optimize the envelope correlation between alpha and beta, not within alpha or beta.

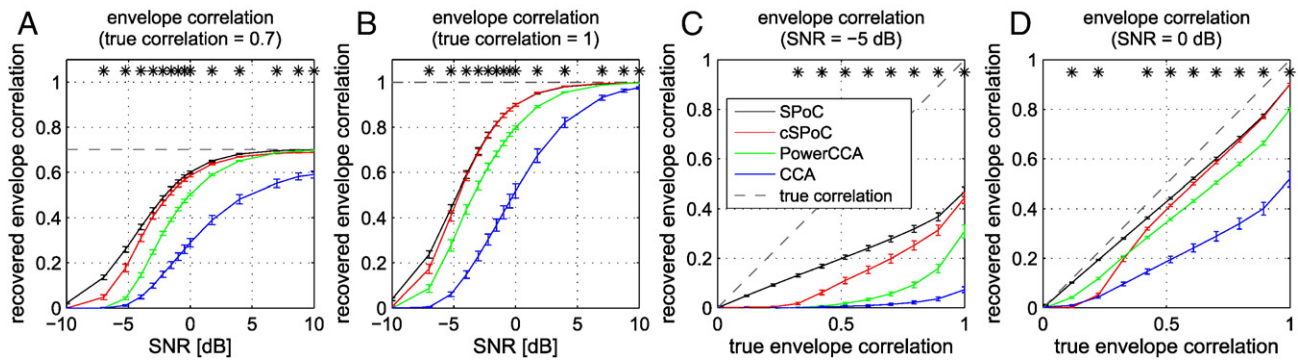


Fig. 3. Simulation results. Empirical correlations between envelopes of source pairs extracted by three unsupervised methods (cSPoC, CCA, PowerCCA) and one supervised method (SPoC). Plots A and B: variation of the SNR for two fixed correlations (0.7 and 1.0, respectively) of the envelopes of the simulated sources (“true” correlations). Plots C and D: variation of the “true” correlation for fixed SNRs (−5 dB and 0 dB, respectively). Empirical correlations were obtained on validation data, which was not used to fit the models. For each parameter setting, simulations were repeated 100 times with each time newly generated model fitting and validation data. Plotted correlations are averaged across repetitions, and errorbars indicate standard error of the mean (SEM). Black asterisks indicate statistically significant differences between cSPoC and PowerCCA (paired t-test, $p < 0.01$).

the figure shows the correlations between the SPoC/cSPoC component envelope and the loudness modulation for each subject (mean over cross-validation folds), as well as the average correlation across subjects, which was 0.64 ± 0.13 (mean $\pm$ standard error, SEM) for SPoC and 0.58 ± 0.17 for cSPoC. Plot B shows the average over subject pairs, which was 0.4 ± 0.11 for SPoC and 0.35 ± 0.16 for cSPoC. Plots C and D show the pairwise envelope correlations for SPoC and cSPoC, respectively (mean over cross-validation folds).

Although having no access to the loudness modulation of the stimulus, cSPoC delivers correlations that are on par with the correlations obtained by SPoC. The gap between the correlations obtained by the two methods widens as the correlation between the envelope of the extracted signal and the target function becomes weaker. This may be caused by lower SNR or lower correlation between the SSAEP component and the loudness modulation. In either case the negative impact on extracted envelope correlations is stronger for cSPoC than for SPoC. This is consistent with observations from the results of the simulations.

The spatial activation patterns of the SSAEP components extracted by each method from each subject are displayed in Fig. 5. The polarity of the SPoC/cSPoC patterns is arbitrary and thus was set such that highest magnitude value is positive. We observe almost no differences in the spatial patterns obtained for individual subjects by cSPoC and SPoC. The spatial patterns show the characteristic maxima (minima) at mid-frontal electrode positions and minima (maxima) at posterior electrode positions, with the sign depending on the phase of the stimulus. Such patterns have previously been reported in event-related potential (ERP) analysis and source modeling of SSAEP responses by Herdman et al. (2002).

Summarizing, the results obtained in this analyses show that cSPoC is able to reliably extract source activity whose envelope dynamics are stimulus induced and consistent across subjects, *without knowledge about the stimulus*.

Identifying generators of sensori-motor rhythms based on envelope anti-correlations in a BCI paradigm

Table 1 contains the classification accuracies (mean over 40 subjects, and standard error) achieved by CSP and cSPoC. The classification results were obtained on validation data, which were not used in the optimization of CSP/cSPoC and LDA. Being a supervised method, CSP achieved the highest accuracy of $71 \pm 2.5\%$ (mean $\pm$ SEM), followed by cSPoC with $68.2 \pm 2.4\%$. Thus the performance of cSPoC is well within range of the performance of CSP, despite the fact that CSP is based on the information about the class labels, while cSPoC has no access to that information.

Fig. 6 shows a comparison of the spatial patterns of CSP and cSPoC components for the three subjects for which highest classification accuracies were achieved using CSP. For each method, the respective components are ordered by their individual class discriminability as indicated by the area under the ROC curve for each component individually. The spatial patterns are ordered from left to right with respect to the discriminability measure, with the most class discriminative being shown on the very left.

While there is a high correspondence between some of the spatial patterns extracted by CSP and cSPoC, they are not identical in all cases. However, both methods extract sources with spatial patterns suggesting

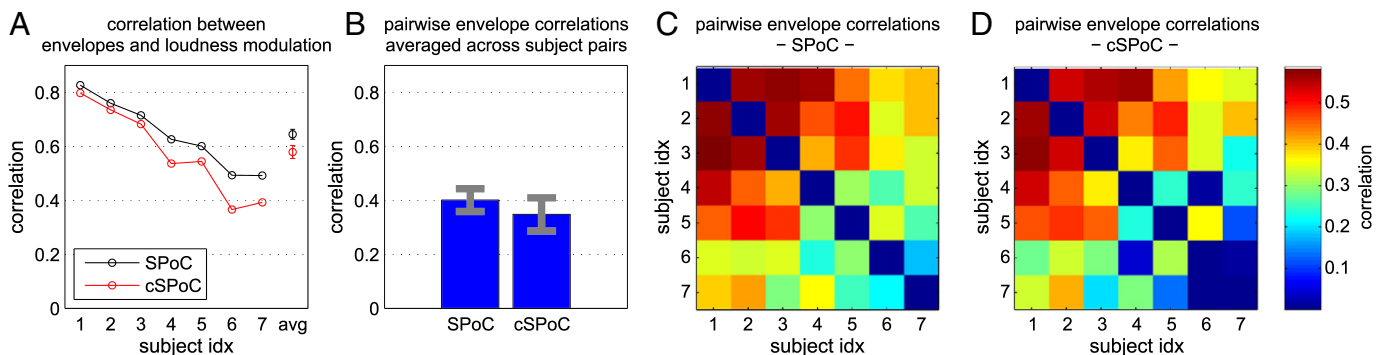


Fig. 4. Results of inter-subject envelope correlation in an auditory steady-state paradigm: correlations. The unsupervised cSPoC method was applied to pairs of subjects, without using the actual loudness modulation. The supervised SPoC method was applied to each subject individually, using its corresponding loudness modulation as a target signal. (A) Cross-validated correlations between envelopes of sources extracted by SPoC/cSPoC and loudness modulation (stimulus intensity) for each subject, as well as averaged across subjects. Errorbars indicate SEM. (B) Pairwise envelope correlations for all subjects as obtained by SPoC and cSPoC, averaged across subject pairs. Errorbars indicate SEM. (C, D) Pairwise envelope correlations for all subjects as obtained by SPoC and cSPoC, respectively.

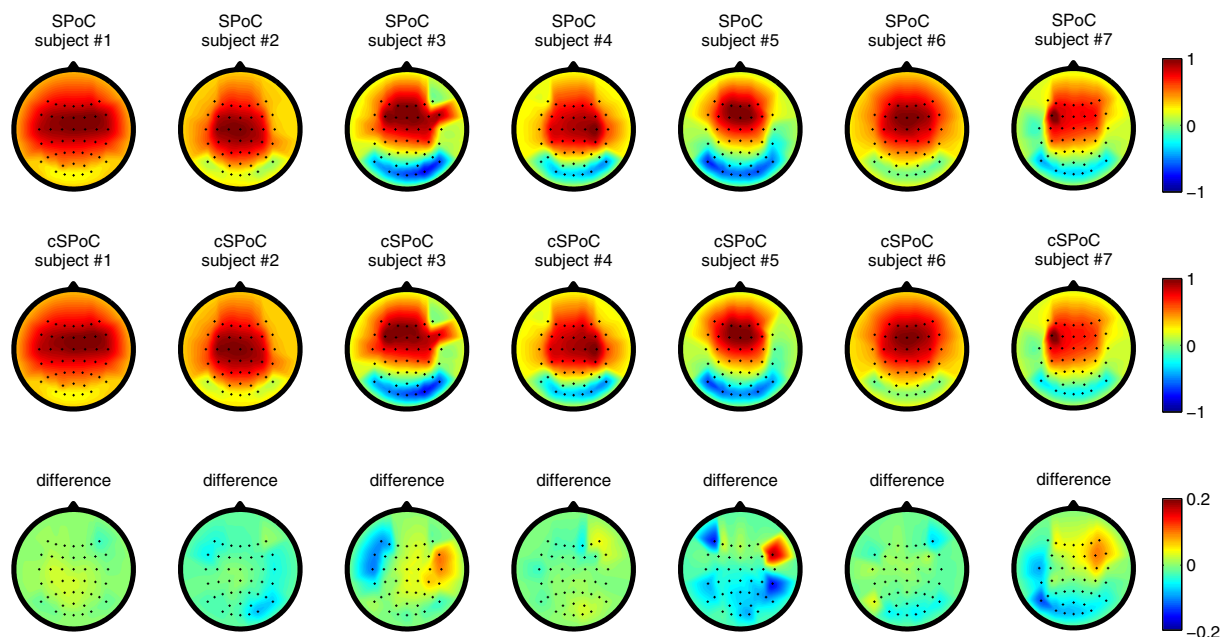


Fig. 5. Results of inter subject envelope correlation in an auditory steady-state paradigm: spatial patterns. For each subject, the top row shows spatial patterns obtained by SPoC, while the second row shows patterns obtained by cSPoC. The third row shows the difference between the patterns of the first and second row. Note that the scaling of the patterns (i.e. magnitude and polarity) is arbitrary in both algorithms. Here the scaling was adjusted such that the largest magnitude was 1. Color scale is the same in rows one and two, but note the different color scale in row three.

a location in motor related areas, as well as parietal and occipital sources. In most subjects, sources with spatial patterns consistent with motor areas are the most task-discriminant. This result is observed not only for well-performing subjects.

Summarizing, our analysis demonstrates that cSPoC is able to extract task-specific sources of ERD activity in an *unsupervised* manner to a degree sufficient to drive a brain–computer interface without notable performance loss compared to using a supervised method. To find these sources with cSPoC, we utilized that the BCI paradigm induces alternating and thereby anti-correlated envelope dynamics of the task-related sensory-motor rhythms.

Analyzing alpha and beta oscillations with coupled envelopes during rest

Fig. 7 shows the cross-validated envelope correlations between the first five cSPoC component pairs, i.e., the correlations between the envelopes of the alpha and beta band sources found by cSPoC. In five out of seven subjects, it can be observed that the first two component pairs deliver significantly higher correlations between alpha and beta compared to the following pairs. For the remaining two subjects (subjects 3 and 4), this is true for the first component pair. Across subjects, the envelope correlations of the first component pair are in the range between 0.4 and 0.8.

The spatial activation patterns of three representative subjects are plotted in Fig. 8. The figure reveals that there is a remarkable similarity of spatial patterns of the alpha and beta band sources within each source pair found by cSPoC, where the similarity is strongest for the component pairs that exhibit highest correlation. This similarity between the spatial patterns indicates that the corresponding neural generators of oscillatory

alpha and beta activity are in very close proximity to each other in the brain, or perhaps even occupy the same cortical space.

In six out of seven subjects, the spatial patterns indicate sensorimotor-related areas as the origins of brain sources with the most strongly coupled envelope dynamics (see subjects 1 and 7 in the figure, for example). However, in subject 5 the spatial patterns suggest sources in parietal or occipital areas to exhibit the largest envelope correlations.

The results of testing within-frequency envelope correlations are shown in Fig. 9. The figure shows the envelope correlations between the first and the second cSPoC alpha (beta) component. There is a statistically significant envelope interaction between the extracted sources in the alpha band in all subjects, as well as for six out of seven subjects in the beta band. Significant correlations range from 0.34 to 0.75 in the alpha band and from 0.41 to 0.6 in the beta band. In all subjects, the envelope correlations between the cSPoC alpha components exceed the envelope correlations between the cSPoC beta components.

Summarizing, the analysis of resting-state data using cSPoC revealed pairs of alpha- and beta band oscillations with large envelope correlations and highly similar physiologically plausible spatial activation patterns. The extracted signals serve as an excellent basis for further investigating the relationship between alpha and beta oscillations, for example by assessing phase synchrony or other measures (see Discussion).

Discussion

In this paper, we have presented cSPoC, a novel unsupervised source separation approach for finding oscillatory sources with envelope correlations. The method was benchmarked using simulations as well as real EEG data. Its performance, which is competitive to supervised approaches, makes cSPoC the method of choice in scenarios in which supervised methods are not applicable, such as hyperscanning settings. It moreover provides a versatile addition to other multivariate analysis tools for cross-frequency coupling such as cross-frequency decomposition (CFD, Nikulin et al., 2012) for phase coupling and multimodal source power correlation (mSPoC, Dähne et al., 2013) for phase-to-power coupling, as well as tools for extracting brain activity from electrophysiological recordings based on other types of dependencies

Table 1

Results of BCI data analysis: classification accuracies. The table shows classification accuracies achieved using CSP (supervised) and cSPoC (unsupervised) as feature extraction methods in a binary motor imagery classification task.

CSP	cSPoC
71 ± 2.5%	68.2 ± 2.4%

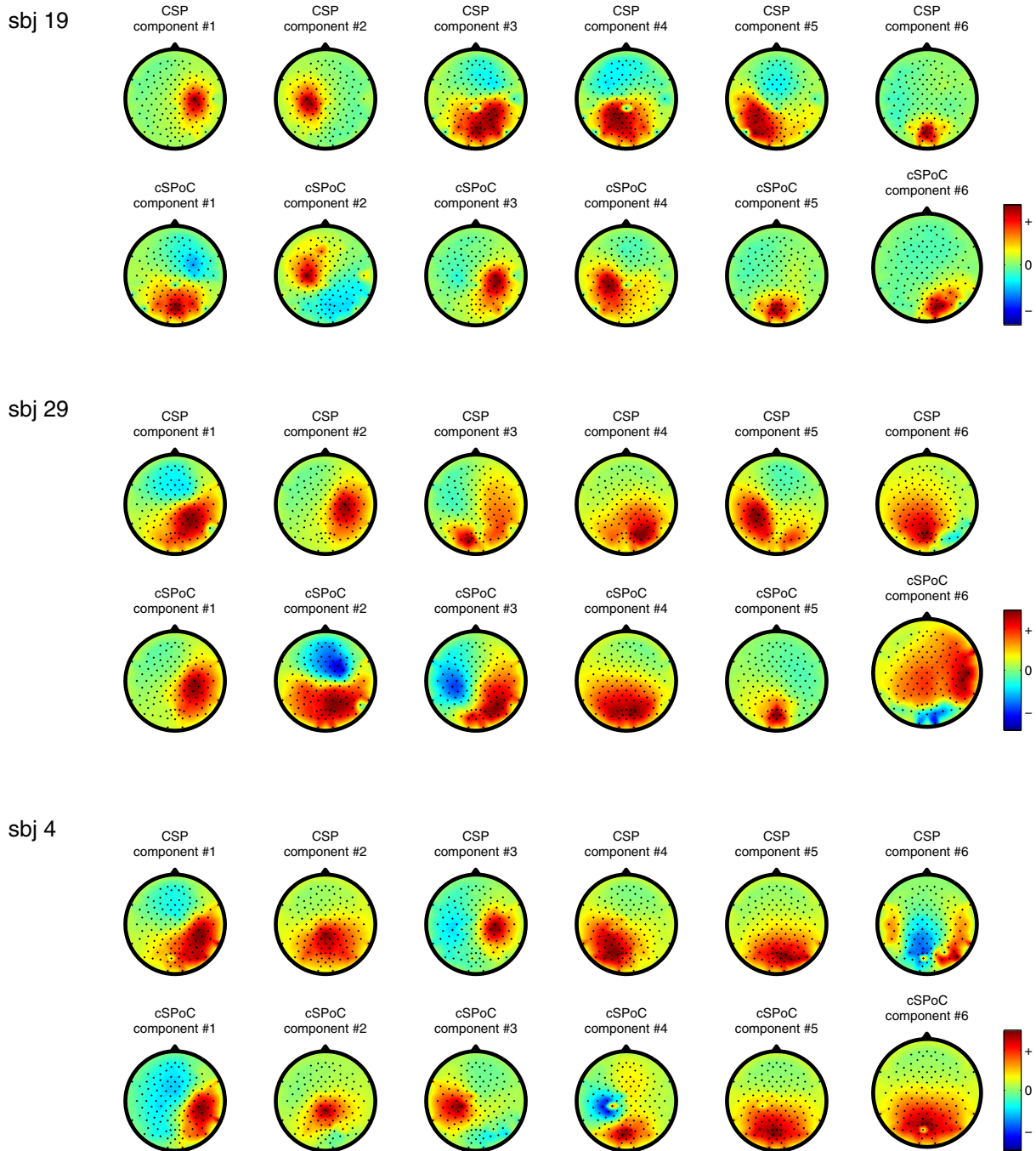


Fig. 6. Results of BCI data analysis: spatial patterns. Spatial patterns of the six CSP/cSPoC components (top/bottom row, respectively) that were used in classification. For each of the three subjects presented, and for each method, the patterns are sorted according to the class-discriminability of the respective envelope features, as assessed by area under the ROC curve. Class-discriminability decreases from left to right.

(Dyrholm et al., 2007; Gómez-Herrero et al., 2008; Haufe et al., 2010; Nolte et al., 2006).

Being a multivariate technique, cSPoC integrates information from all recording channels and thereby achieves higher SNRs compared to single channel analysis, which leads to stronger effects. By projecting the data into the cSPoC source space, the dimensionality is reduced and thus the problem of channel-wise comparison/correlations (and multiple testing) is avoided. In our examples, we demonstrated the robustness of the obtained envelope correlations by means of cross-validation and permutation tests. While statistical approaches have been developed to assess functional connectivity after projection into a source space (e.g. Carbonell et al., 2009), these approaches typically test the connectivity of source time-courses, rather than connectivity of their envelopes.

Thus, it is desirable to extend such statistical models in order to avoid potentially time-consuming permutation analysis.

Related methods

Canonical Source Power Analysis is conceptually and structurally related to a number of existing methods. Among those are two recent approaches also aiming to extract brain oscillations with power dependencies, PowerCCA by Ramírez et al. (2013) and non-linear CCA by Campi et al. (2013). We included PowerCCA in the simulations. The reference to the respective (sub) section seems overly complicated now. We are ok with simply removing the reference to the sub section, where it was outperformed by cSPoC. We believe the reason for that

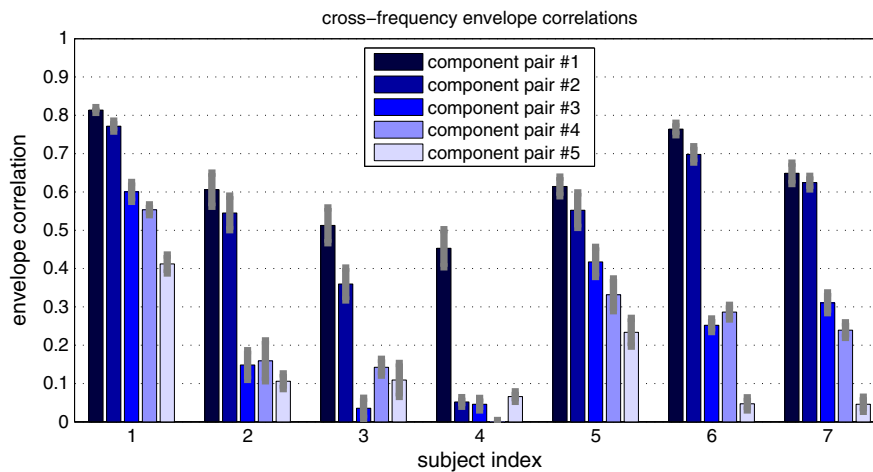


Fig. 7. Results of resting state alpha/beta analysis: cross-frequency envelope correlations. Cross-validated correlations between the envelopes of alpha and beta oscillations of the first five cSPoC component pairs for each subject. See Fig. 8 for corresponding spatial patterns of selected subjects. Errorbars indicate SEM over cross-validation folds.

to be the different approaches for optimizing envelope correlations employed by cSPoC and PowerCCA. PowerCCA optimizes over huge $N_{x,y}^2$ -dimensional parameters matrices, where N_x and N_y are the number of recording channels for data sets $\mathbf{x}$ and $\mathbf{y}$. The actual extraction filters are obtained post-hoc as the first eigenvectors of these matrices. In contrast, cSPoC optimizes over filter vectors directly and has thus only $N_x + N_y$ parameters to tune. For this reason, cSPoC may be less prone to overfitting than PowerCCA. Non-linear CCA by Campi et al. (2013), which is related to an earlier approach by Gutmann and Hyvärinen (2011), essentially computes linear CCA in the space of all binomials of the original input space, with a dimensionality constraint on the weight vectors. While their method achieves good results on combined data of 2×5 subjects, it failed to do so on single subject-pair level. A crucial difference between nonlinear CCA and cSPoC is that cSPoC acts on the analytic signal and is thereby able to optimize envelope correlations directly, whereas nonlinear CCA uses the (temporally smoothed) squared signal as a proxy for the instantaneous power.

There is also an interesting relation between cSPoC and algorithms that optimize higher order (cross-) moments of the projections. Among these algorithms are a specific class of independent component analysis (ICA) algorithms (Cardoso and Souloumiac, 1993; Comon, 1994; Hyvärinen and Oja, 1997), as well as minimum overlap component analysis (MOCA) (Nolte et al., 2009). These algorithms may arrive at similar solutions using an entirely different set of assumptions and constraints. Yet, they are not explicitly designed to optimize envelope correlations of projections of multivariate oscillatory signals.

Implications of experimental results

It is usually believed that the amplitude of local field potential (LFP) and EEG signals represents spatially averaged synaptic activity. Furthermore, a stronger synchronization between neurons relates to larger amplitude in LFP/EEG signals (Denker et al., 2011; Pfurtscheller and Lopes da Silva, 1999). The correlation between the envelopes of alpha and beta oscillations found in the present study indicates that changes in the strength of local synchronization among the neurons generating alpha oscillations are accompanied by similar changes in the strength of local synchronization among the neurons generating beta oscillations. Such power-to-power coupling between alpha and beta oscillations might be important for the coordination of cortical excitability changes, reflected primarily in the amplitude of alpha oscillations, with the dynamics of beta oscillations, known to be involved in different motor sensori-motor tasks.

In addition, a different interpretation of our results is possible, which was discussed in Nikulin and Brismar (2006). Periodic but not strictly

sinusoidal post-synaptic activity might result in oscillations with noticeable spectral energy peaks at harmonic frequencies. This implies that beta oscillations could be the first harmonic of a quasi-sinusoidal process with alpha base frequencies. Specifically, this theory not only implies large envelope correlations between alpha and beta, it also predicts that the spatial patterns associated with alpha-synchronous beta activity should match the spatial patterns of the corresponding alpha frequency process. In our experiment, we observed matching spatial patterns for maximally envelope-correlated alpha and beta signals.

Our results could be a stepping stone for further investigation into the relationship of alpha and beta oscillations during rest. Specifically, it may provide a means of separating alpha-harmonic beta oscillations from genuine neural processes that manifest themselves in beta activity only.

In our auditory experiments, we showed that by using cSPoC one can obtain spatial filters that extract neuronal activity, which correlates across different subjects. While for those experiments we also had a common target function (acoustic stimulus), other scenarios are conceivable where such target functions are not available and yet there is a need to estimate a similarity between neuronal activation across the subjects. For instance in studies in problem solving or memory encoding/retrieval one can analyze a common pattern of neuronal activity from the beginning of the task or stimulus. In this case a target function is not available, yet many subjects can present a typical pattern of neuronal activations which can be extracted with cSPoC. Moreover, by studying common patterns (obtained with cSPoC) as well as residual patterns (after the subtraction of cSPoC related data) one can also investigate neuronal processes related to each specific individual. This in turn might allow to investigate one of the most persistent problems in cognitive neuroscience, namely the large variability of neuronal patterns across subjects even in the same experimental task.

Clinical perspective

cSPoC can also be used in clinical studies. For instance recent studies in patients with Parkinson disease revealed an existence of phase-synchronization in beta frequency range between subthalamic nucleus (STN) and cortex (Klostermann et al., 2007; Litvak et al., 2011). Here an interesting question would be how the envelope of alpha oscillations (reflecting cortical excitability) might affect local neuronal dynamics in STN, the latter being expressed in the amplitude of beta oscillations. These alpha versus beta envelope interactions can be studied with cSPoC.

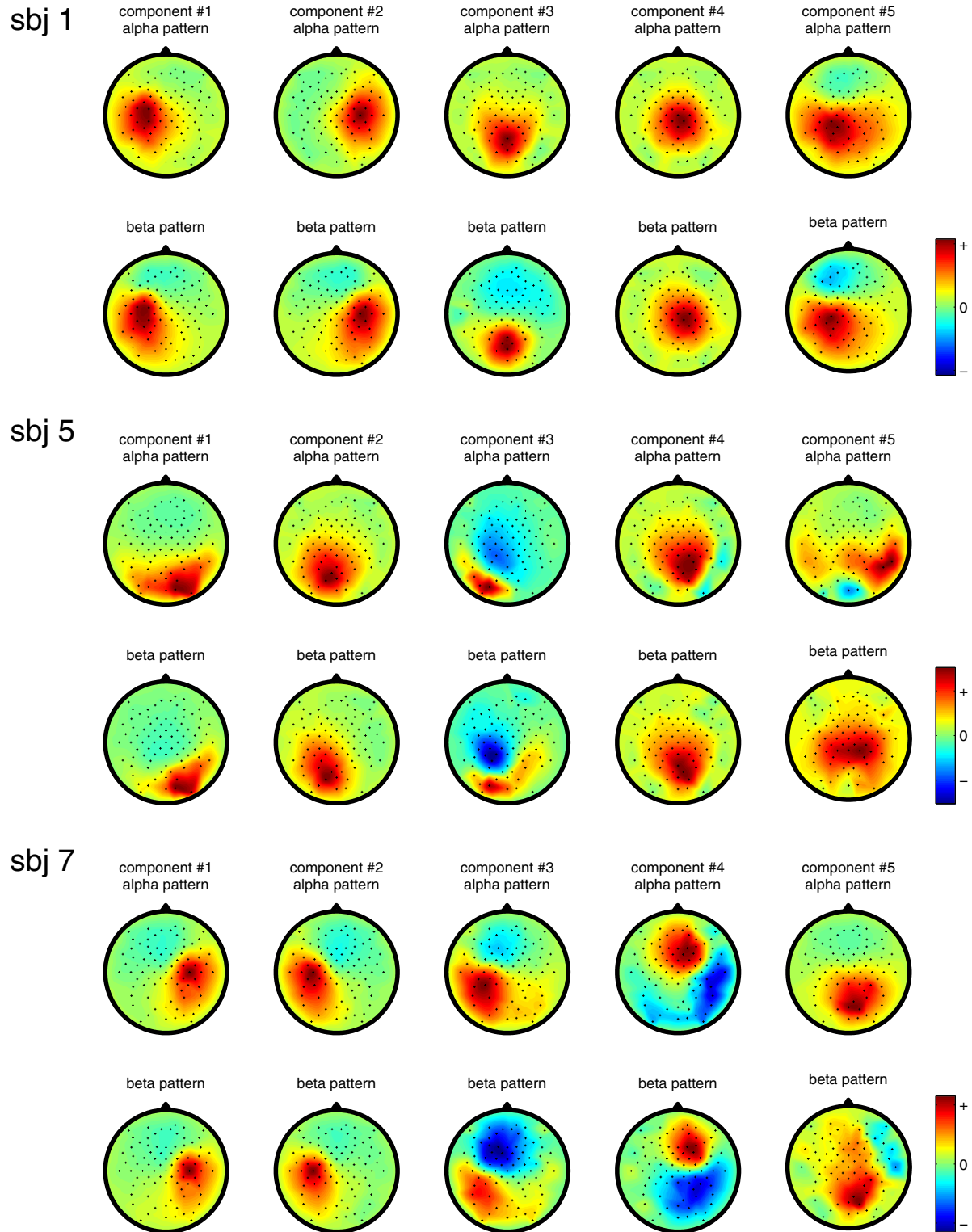


Fig. 8. Results of resting state alpha/beta analysis: spatial patterns. Spatial activation patterns of the five cSPoC components pairs with best correlating alpha/beta envelopes. For each of the three subjects presented, the top row shows the spatial patterns of components in the alpha band while the bottom row shows the respective best correlating component in the beta band. The envelope correlations between the alpha/beta pairs can be found in Fig. 7 and the envelope correlations between the first two alpha components and the first two beta components can be found in Fig. 9.

Conclusion

We presented a novel method, cSPoC, which is able to identify brain oscillations with correlated envelopes from high-dimensional and noisy

electrophysiological recordings. The method performs excellently in simulations and showed very good results in three distinctively different, yet highly relevant, analyses scenarios involving real EEG data. We believe that cSPoC will be a valuable tool in the quest for

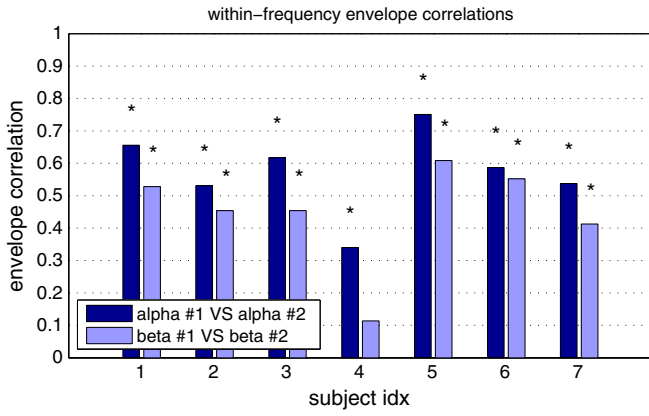


Fig. 9. Results of resting state alpha/beta analysis: within-frequency envelope correlations. Correlations were computed between envelopes of the first two cSPoC component pairs, separately for alpha and the beta band. See Fig. 8 for corresponding spatial patterns of selected subjects. Asterisks indicate statistical significance with $p < 0.001$, permutation test.

understanding the mechanisms and functions of power-to-power coupling in electrophysiological recordings.

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Appendix A. Gradients

Here we provide the gradient of the objective function Eq. (6) with respect to $\mathbf{w}_x$. For symmetry reasons, the gradient with respect to $\mathbf{w}_y$ is obtained analogously.

$$\frac{\partial \text{Corr}[\Phi_x, \Phi_y]}{\partial \mathbf{w}_x} = \frac{\frac{\partial \langle \Phi_x \Phi_y \rangle}{\sqrt{\langle \Phi_x^2 \rangle \langle \Phi_y^2 \rangle}}}{\frac{\partial \mathbf{w}_x}} = \frac{\langle \Delta_{yx} \frac{\partial \Phi_x}{\partial \mathbf{w}_x} \rangle}{\sqrt{\langle \Phi_x^2 \rangle \langle \Phi_y^2 \rangle}}, \quad (\text{A.1})$$

with

$$\bar{\Phi}_i = \Phi_i - \langle \Phi_i \rangle, \quad (\text{A.2})$$

for $i \in \{x, y\}$,

$$\Delta_{yx} := \bar{\Phi}_y - \bar{\Phi}_x \frac{\langle \bar{\Phi}_x \bar{\Phi}_y \rangle}{\langle \bar{\Phi}_x^2 \rangle}, \quad (\text{A.3})$$

$$\frac{\partial \bar{\Phi}_x}{\partial \mathbf{w}_x} = \frac{\partial \Phi_x}{\partial \mathbf{w}_x} - \left\langle \frac{\partial \Phi_x}{\partial \mathbf{w}_x} \right\rangle. \quad (\text{A.4})$$

For $\Phi_x = \sqrt{(\mathbf{w}_x^\top \mathbf{x})^2 + (\mathbf{w}_x^\top \mathbf{H}[\mathbf{x}])^2}$, we have

$$\frac{\partial \Phi_x}{\partial \mathbf{w}_x} = \frac{1}{\Phi_x} \left((\mathbf{w}_x^\top \mathbf{x}) \mathbf{x} + (\mathbf{w}_x^\top \mathbf{H}[\mathbf{x}]) \mathbf{H}[\mathbf{x}] \right), \quad (\text{A.5})$$

with $\mathbf{H}[\mathbf{x}]$ denoting the Hilbert transform of $\mathbf{x}$.

Appendix B. Extensions

Here, we provide a general formalism for integrating the extensions proposed in the section [Canonical source power correlation analysis \(cSPoC\)](#). To this end, let us consider to optimize correlations between general functions f of Φ_x and Φ_y , where f must be differentiable. The optimization problem becomes

$$\max_{\mathbf{w}_x, \mathbf{w}_y} c_{sg} \cdot \text{Corr}[f(\Phi_x), f(\Phi_y)]. \quad (\text{B.1})$$

The partial derivative of this problem with respect to $\mathbf{w}_x$ is given by

$$\frac{\partial \text{Corr}[f(\Phi_x), f(\Phi_y)]}{\partial \mathbf{w}_x} = \frac{\frac{\partial \langle \bar{f}(\Phi_x) \bar{f}(\Phi_y) \rangle}{\sqrt{\langle \bar{f}(\Phi_x)^2 \rangle \langle \bar{f}(\Phi_y)^2 \rangle}}}{\frac{\partial \mathbf{w}_x}} = \frac{\langle \Delta_{yx} \frac{\partial \bar{f}(\Phi_x)}{\partial \mathbf{w}_x} \rangle}{\sqrt{\langle \bar{f}(\Phi_x)^2 \rangle \langle \bar{f}(\Phi_y)^2 \rangle}} \quad (\text{B.2})$$

with

$$\bar{f}(\Phi_i) := f(\Phi_i) - \langle f(\Phi_i) \rangle, \quad \text{for } i \in \{x, y\}, \quad (\text{B.3})$$

$$\Delta_{yx} := \bar{f}(\Phi_y) - \bar{f}(\Phi_x) \frac{\langle \bar{f}(\Phi_x) \bar{f}(\Phi_y) \rangle}{\langle \bar{f}(\Phi_x)^2 \rangle}, \quad (\text{B.4})$$

$$\frac{\partial \bar{f}(\Phi_x)}{\partial \mathbf{w}_x} = \frac{\partial f(\Phi_x)}{\partial \mathbf{w}_x} - \left\langle \frac{\partial f(\Phi_x)}{\partial \mathbf{w}_x} \right\rangle. \quad (\text{B.5})$$

Finally, we have

$$\frac{\partial f(\Phi_x)}{\partial \mathbf{w}_x} = \frac{\partial f(\Phi_x)}{\partial \Phi_x} \cdot \frac{\partial \Phi_x}{\partial \mathbf{w}_x}, \quad (\text{B.6})$$

where $\frac{\partial \Phi_x}{\partial \mathbf{w}_x}$ is given in Eq. (A.5) and $\frac{\partial f(\Phi_x)}{\partial \Phi_x}$ depends on the specific choice of the function f and will be given below for some examples.

Logarithm

Let f be the natural logarithm, then we have

$$f_{\log}(\Phi_x) := \log(\Phi_x), \quad (\text{B.7})$$

and

$$\frac{\partial f_{\log}(\Phi_x)}{\partial \Phi_x} = \frac{1}{\Phi_x}. \quad (\text{B.8})$$

Average within epochs

In some scenarios it might be beneficial to average the envelope of the projection within epochs or trials that are denoted by the index e . Then we correlate $f(\Phi_{\mathbf{x}})$ and $f(\Phi_{\mathbf{y}})$ across the epoch index e , rather than across the original time index t . Let T_e be the set of all time indices t that are in the epoch e . We define

$$f_e(\Phi_{\mathbf{x}}) := \frac{1}{|T_e|} \sum_{t \in T_e} \Phi(t), \quad (\text{B.9})$$

where $|T_e|$ denotes the cardinality of the set T_e . Note that f_e now has a sampling rate of epochs, rather than the original time index t . The corresponding partial derivative is given by

$$\frac{\partial f_e(\Phi_{\mathbf{x}})}{\partial \mathbf{w}_{\mathbf{x}}} = \frac{1}{|T_e|} \sum_{t \in T_e} \frac{\partial \Phi(t)}{\partial \mathbf{w}_{\mathbf{x}}}. \quad (\text{B.10})$$

Time-lagged envelope coupling

Time-lagged interactions between envelopes can be modeled by introducing a temporal filtering function $\mathbf{w}_{\tau} \in \mathbb{R}^{N_{\tau}}$ to one of the envelope functions, with N_{τ} being the number of time-lags considered. Let $\{\tau_i\}$, with $i \in \{1, \dots, N_{\tau}\}$, denote a list of suitable time (or epoch) indices. Then we define

$$f_{\tau}(\Phi_{\mathbf{x}})(t) := \sum_i^{N_{\tau}} (\mathbf{w}_{\tau})_i \Phi_{\mathbf{x}}(t - \tau_i), \quad (\text{B.11})$$

where $(\mathbf{w}_{\tau})_i$ denotes the i th element of the vector $\mathbf{w}_{\tau}$. In this formulation, $\mathbf{w}_{\tau}$ becomes a parameter vector over which one can optimize in the same way as over $\mathbf{w}_{\mathbf{x}}$ and $\mathbf{w}_{\mathbf{y}}$. The partial derivatives for this extension are

$$\frac{\partial f_{\tau}(\Phi_{\mathbf{x}})}{\partial \mathbf{w}_{\mathbf{x}}} = \sum_i^{N_{\tau}} (\mathbf{w}_{\tau})_i \frac{\partial \Phi_{\mathbf{x}}(t - \tau_i)}{\partial \mathbf{w}_{\mathbf{x}}} \quad (\text{B.12})$$

$$\frac{\partial f_{\tau}(\Phi_{\mathbf{x}})}{\partial (\mathbf{w}_{\tau})_i} = \Phi_{\mathbf{x}}(t - \tau_i). \quad (\text{B.13})$$

Note that envelopes vary on a slower time scale than the carrier signals. Thus it is useful to combine the last two proposed extensions such that during the optimization $\Phi_{\mathbf{x}}$ is downsampled by averaging within short time intervals before applying the temporal filter $\mathbf{w}_{\tau}$. This greatly reduces the number of coefficients in $\mathbf{w}_{\tau}$.

Appendix C. Simulated EEG

The following equations describe the generation of the pseudo-EEG measurement $\mathbf{x}(t)$ as the sum of contributions $\mathbf{x}_s(t)$ due to target sources, contributions $\mathbf{x}_{ba}(t)$ due to background sources, and sensor noise $\mathbf{x}_{noise}(t)$:

$$\mathbf{x}_s(t) = \sum_{i=1}^K \mathbf{a}_{\mathbf{x}}^i(\mathbf{s}_{\mathbf{x}}(t))_i \quad (\text{C.1})$$

$$\mathbf{x}_{ba}(t) = \sum_{i=K+1}^{K_s} \mathbf{a}_{\mathbf{x}}^i(\mathbf{s}_{\mathbf{x}}(t))_i \quad (\text{C.2})$$

$$\mathbf{x}_{noise}(t) = \frac{1}{c_{ba}} \mathbf{x}_{ba}(t) + \gamma_e \epsilon_{\mathbf{x}}(t) \quad (\text{C.3})$$

$$\mathbf{x}(t) = \gamma \frac{\mathbf{x}_s(t)}{c_s} + \frac{\mathbf{x}_{noise}(t)}{c_{noise}}. \quad (\text{C.4})$$

The constants c_{ba} , c_s , and c_{noise} are the Frobenius norms of the corresponding data matrices $\mathbf{x}_{ba}$, $\mathbf{x}_s$, and $\mathbf{x}_{noise}$, and serve to balance the data matrices prior to building the weighted sums. For the matrix $\mathbf{x}_{(\cdot)}$, the normalized Frobenius norm corresponds to

$$c_{(\cdot)} = \frac{1}{N_x T} \sum_i^{N_k} \sum_t^T (\mathbf{x}_{(\cdot)}(t))_i^2, \quad (\text{C.5})$$

i.e., it measures the signal energy, averaged across channels and time. The scaling parameter γ_e determines the strength of (Frobenius-) normalized Gaussian noise $\epsilon_{\mathbf{x}}$, and was set to 0.1 in all simulations. The scaling parameter $\gamma > 0$ determines the ratio between the energy of the scalp-projected target source time courses and the energy of all other (background and sensor noise) contributions in the data. Thus the signal-to-noise ratio in dB scale is given by $\text{SNR} = 10 \log_{10}(\gamma)$.

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